

Applications: Growth and Decay

ANSWER KEY



Growth models

- 1 A city's population is modeled by $P(t) = 12000(1.035)^t$, where t is in years.
- a State the initial population and the annual growth rate. Initial population 12,000; growth rate 3.5% per year.
- b Find the population after 10 years, to the nearest person. $P(10) = 12000(1.035)^{10} \approx 16,927$.
- 2 An account earns 4% annual interest compounded continuously. How long does it take an investment to double? Give an exact expression and a decimal to two places.
- $2 = e^{0.04t}$, so $t = \frac{\ln 2}{0.04} \approx 17.33$ years.
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Decay models

- 3 Carbon-14 has a half-life of 5730 years: $A(t) = A_0\left(\frac{1}{2}\right)^{t/5730}$.
- a What fraction of the original carbon-14 remains after 2000 years? Answer to three decimal places.
- $\left(\frac{1}{2}\right)^{2000/5730} \approx 0.785$.
- b A sample retains 30% of its carbon-14. Estimate its age to the nearest hundred years.
- $0.3 = (0.5)^{t/5730}$ gives $t = 5730 \frac{\ln 0.3}{\ln 0.5} \approx 10,000$ years.
- 4 A 200 g sample of a radioactive substance decays to 50 g in 12 hours, following $A(t) = 200e^{kt}$.
- a Find the decay constant k (4 d.p.). $\frac{50}{200} = e^{12k}$, so $k = \frac{\ln(1/4)}{12} \approx -0.1155$.
- b Find the half-life of the substance. $200 \rightarrow 50$ is two halvings in 12 h, so the half-life is exactly 6 hours (equivalently $\frac{\ln 2}{0.1155} \approx 6.0$ h).
- 5 Newton's law of cooling: coffee at 90°C cools in a 20°C room following $T(t) = 20 + 70e^{-0.1t}$ (t in minutes). Find the temperature after 10 minutes, to one decimal place.
- $T(10) = 20 + 70e^{-1} \approx 20 + 25.75 = 45.8^\circ\text{C}$.
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Logistic growth (an introduction)

- 6 A lake is stocked with fish, and the population is modeled by $P(t) = \frac{5000}{1 + 9e^{-0.4t}}$, where t is in years.
- a Find the initial population $P(0)$. $P(0) = \frac{5000}{1+9} = 500$.
- b Find the carrying capacity (the value P approaches as $t \rightarrow \infty$). As $t \rightarrow \infty$, $e^{-0.4t} \rightarrow 0$, so $P \rightarrow \frac{5000}{1} = 5000$.
- c Find $P(10)$, to the nearest fish. $P(10) = \frac{5000}{1 + 9e^{-4}} \approx \frac{5000}{1.1648} \approx 4292$.

Comparing models

- 7 Two towns start with 10,000 people. Town A grows at a constant rate of 300 people/year (linear); Town B grows at 3% per year (exponential). Find each town's population after 15 years, to the nearest person, and state which is larger.
Town A: $10000 + 300(15) = 14,500$. Town B: $10000(1.03)^{15} \approx 10000(1.5580) \approx 15,580$. Town B is larger.
- 8 A radioactive isotope decays from 80 g to 10 g over 3 half-lives. Find the half-life in terms of the decay time, and write the exponential decay model $A(t) = 80e^{kt}$, finding k if the total elapsed time is 15 years (4 d.p.).
 $80 \rightarrow 10$ over 3 halvings ($80 \rightarrow 40 \rightarrow 20 \rightarrow 10$) means 3 half-lives = 15 years, so half-life = 5 years.
Using $A(15) = 10$: $10 = 80e^{15k}$, so $k = \frac{\ln(1/8)}{15} \approx -0.1386$.
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Visualizing the logistic curve

- 9 The fish population model $P(t) = \frac{5000}{1 + 9e^{-0.4t}}$ (from earlier) is graphed below. Use the graph to estimate where the population is growing fastest, then explain why this happens near the inflection point.
The curve is steepest near its midpoint, roughly where $P \approx 2500$ (half the carrying capacity), consistent with the logistic model's fastest growth occurring at $P = \frac{L}{2}$.
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Doubling time and half-life formulas

- 10 Derive a general formula for the doubling time T of a quantity growing as $A(t) = A_0e^{kt}$, in terms of k .
Set $2A_0 = A_0e^{kT}$: $2 = e^{kT} \Rightarrow \ln 2 = kT \Rightarrow T = \frac{\ln 2}{k}$.
- 11 Using the doubling-time formula derived above, find the doubling time for a population growing at a continuous rate of $k = 0.025$ per year, to one decimal place.
 $T = \frac{\ln 2}{0.025} \approx \frac{0.6931}{0.025} \approx 27.7$ years.
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Half-life formula derivation

- 12 Derive a general formula for the half-life $T_{1/2}$ of a quantity decaying as $A(t) = A_0e^{kt}$ (with $k < 0$), in terms of k .
Set $\frac{A_0}{2} = A_0e^{kT_{1/2}}$: $\frac{1}{2} = e^{kT_{1/2}} \Rightarrow \ln\left(\frac{1}{2}\right) = kT_{1/2} \Rightarrow T_{1/2} = \frac{-\ln 2}{k}$ (positive since $k < 0$).
- 13 A radioactive isotope has decay constant $k = -0.023$ per year. Use the formula above to find its half-life, to one decimal place.
 $T_{1/2} = \frac{-\ln 2}{-0.023} = \frac{0.6931}{0.023} \approx 30.1$ years.

Combining growth models with rates

- 14** A savings account earns continuous interest, growing from 5000 to 5512.50 riyals in 2 years. Find the annual interest rate r using $A = Pe^{rt}$, to two decimal places (as a percentage).

$5512.50 = 5000e^{2r} \Rightarrow 1.1025 = e^{2r} \Rightarrow 2r = \ln(1.1025) \approx 0.0976 \Rightarrow r \approx 0.0488$, i.e. approximately 4.88%.